

Intelligent Grid Network for Transformer Health Monitoring, Life Prediction, and Environmental-Risk-Aware Maintenance Using Edge AI and LoRa Telemetry

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Abstract—The distribution transformer is continuously subjected to electrical, thermal, mechanical and environmental stresses and the traditional maintenance method is still mainly manual, periodic and reactive, and some may be too expensive for a small or remote substation to adopt such as continuous dissolved-gas analysis and SCADA [1] [2]. In this paper, the development of the low-cost transformer health and life-prediction system, based on Electrical Stress Fingerprinting (ESF) technique, into a decentralized Intelligent Grid Network [13] is presented. In the new architecture, the original edge node is expanded to collect voltage, load current, temperature, oil level and vibration, together with the ambient environment parameters, and calculate the normalized stress indicators, such as a 0-100 Transformer Health Index (THI) and a degradation-based remaining useful life (RUL) estimation locally [3], [7], [13]. The edge node combines transformer condition with environmental-risk factors like ambient temperature, relative humidity and fire receptivity due to wind, and sends compact, json-formatted telemetry data to a local host through a point-to-point LoRa (SX1278) link, without any cellular subscription or cloud-gateway required. Under normal conditions, the Python/Streamlit dashboard displays real-time gauges, trend plots and the dynamically projected next-service date, and, under anomalous conditions, invokes an offline, locally-hosted, large-language-model (LLM) layer with retrieval-augmented technical grounding, to produce a maintenance-oriented diagnostic explanation and technician playbook [6]. Protection in the form of threshold alarms and relay disconnection remains deterministic and is located solely on the embedded edge controller, allowing the generative AI layer to only serve as a decision support tool without being a protection tool [5] [6]. Unvalidated performance figures are not introduced and only validated prototype capabilities of the environment risk fusion and LoRa communication system and the dashboard and offline AI-diagnostic capabilities are reported, as these were carried over from the baseline ESF system design. The split-intelligence design is designed to be deployable in the decentralized and rural grid infrastructure, which are in low costs. The split-intelligence

design is expected to be suitable for low-cost, decentralized deployment across rural and distributed grid infrastructure [1], [3], [8], [13].

Index Terms—Transformer Health Monitoring, Electrical Stress Fingerprinting, Transformer Health Index, Remaining Useful Life, Internet of Things, Edge Computing, LoRa Communication, Environmental Risk Fusion, Edge AI, Retrieval Augmented Generation, Predictive Maintenance, STM32.

I. INTRODUCTION

Distribution transformers are important parts of electrical distribution networks and they work all the time under different kinds of stresses like electricity, heat, movement and the environment. Things like much current changes in voltage getting too hot, shaking and problems with oil levels can slowly make the transformer worse and cause it to fail and stop working [1] [3] [8]. Traditional ways of taking care of the transformers are usually only done when something goes wrong or on a schedule. Fixed rules for protection only react when things are really bad not when they are slowly getting worse. Better methods like SCADA and dissolved-gas analysis give a lot of information about how the transformer is doing but setting them up can be too expensive for small or remote stations [1] [2].

IoT devices and smart monitoring systems that work close to the source offer a way. These devices measure things like voltage, current heat, oil level and shaking. They also figure out how healthy the transformer is [3] [4] [8]. A previous step in this work made a low-cost system that uses an STM32 chip to check the health of a transformer and predict how long it will last. This system used Electrical Stress Fingerprinting to make a Transformer Health Index and gave warnings and protection [13]. The original system worked mostly on its own and did not include the environment or let people know about the condition from far away.

This work adds features to the ESF/THI system. It makes a network of parts by adding sensors for environmental risks and combining those risks at the edge. It uses LoRa for sending data between points a Python and Streamlit dashboard for monitoring predicts when the transformer may need service based on how it's getting worse and adds a part that uses AI to explain what is happening and help make decisions. It combines the health of the transformer with things like temperature and humidity to give maintenance advice. The old alarms and protections still work on the STM32 device while the AI part is only used to explain and help decide what to do

The main thing this work adds is a way to split the work between the edge and the main system. This makes the tested ESF/THI system better for places that need a low-cost way to monitor transformers in areas [7] [13]. The rest of the paper talks about the problem what others have done the new design, how the sensors and ESF/THI work, predicting how long the transformer will last, combining with risks sending data using LoRa using AI for diagnosis making the dashboard testing the system comparing it with other methods what is not good, about it what can be done next and what we have learned.

II. PROBLEM STATEMENT AND RESEARCH GAPS

A. Gap 1 — Limited Adaptive Health Intelligence

Current transformer monitoring devices analyze data points separately, failing to provide a complete health picture [1], [3], [8]. The old ESF/THI system addressed this by combining voltage, current, temperature, and vibration into a single health score [13]. Our new system continues this approach and adds a feature to track the rate of degradation, as detailed in Section VII [7], [13].

B. Gap 2 — Environmental-Risk Isolation

Transformer health assessments often overlook environmental conditions. Separate monitoring equipment and weather data prevent understanding how factors like heat, dryness, or wind increase failure risk [12]. Our new system incorporates an "edge risk-fusion layer" (Section VIII) to merge internal transformer data with external environmental information.

C. Gap 3 — Communication and Diagnostic Limitations

Existing internet-connected monitoring systems can be costly and unreliable, especially in rural areas [1], [2]. Smaller devices often provide only basic alerts without clear explanations [3], [8]. Our new system uses LoRa technology for internet-independent communication (Section IX) and includes a diagnostic tool that can identify and explain issues for maintenance purposes without an internet connection [6].

III. RELATED WORK

Kumar et al. presented an IoT-based transformer monitoring system for continuous data acquisition and remote fault analysis [1]. Hashemi and Dikmen combined online DGA with electrical and thermal measurements for early fault detection [2]. Ahmad et al. demonstrated machine-learning-based transformer health monitoring using IoT edge computing [3]. Hasan et al. developed

an IoT-based smart transformer with real-time temperature, humidity and load monitoring and self-protection [4].

Bainy and Silva demonstrated deterministic differential protection methods for power systems [5]. Bazi et al. applied AI for transformer fault diagnosis [6], while Sudrajat et al. used ANN-based prediction of transformer health index [7]. Akther et al. developed IoT-based transformer health monitoring with wireless communication [8]. Asutkar et al. investigated health-index-based transformer evaluation [9], Tonoy proposed IoT-based predictive maintenance [10], and Shetty et al. presented cloud-integrated fault detection and alerting [11].

Byrne and Patrick established the relevance of temperature and humidity for wildfire-risk assessment [12]. The earlier work of the present project developed the STM32-based Electrical Stress Fingerprinting transformer health and life-prediction system that forms the basis of this work [13].

Overall, existing studies demonstrate transformer sensing, health indexing, IoT/edge monitoring, and AI-based diagnosis, but do not combine transformer health, localized environmental-risk fusion, gateway-free LoRa communication, and offline RAG-based AI diagnostics in a single low-cost architecture, which is the focus of this work.

IV. PROPOSED SYSTEM ARCHITECTURE

The proposed Intelligent Grid Network is organized as two cooperating layers: a decentralized edge node at the transformer site, and an intelligent gateway/host terminal operated by the maintenance team. Fig. 1 shows the split-intelligence architecture. Deterministic, safety-relevant computation, sensing, normalization, THI calculation, threshold evaluation and relay protection are kept entirely on the edge controller, while visualization, historical trend analysis, service-date projection and generative diagnostic reasoning are performed at the host. This partitioning ensures that the transformer remains protected even if the LoRa link, host terminal or AI service is temporarily unavailable.

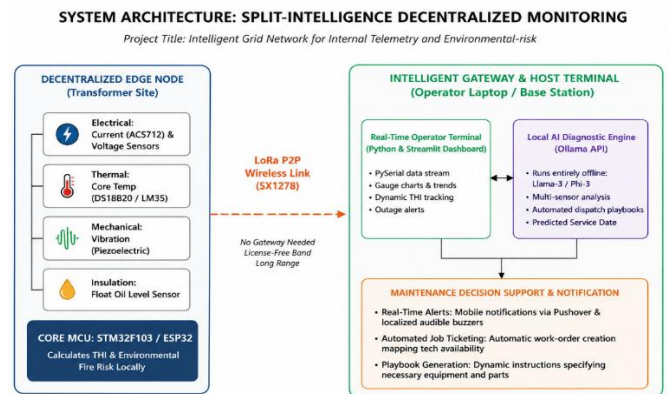


Fig. 1. Proposed decentralized edge-host (split-intelligence) architecture for transformer health and environmental-risk monitoring.

A. Decentralized Edge Node

The edge node uses the same STM32 setup for sensing and protection as the original ESF system but adds environmental sensing [13]. It measures voltage and current for electrical strain,

temperature for heat stress, vibration for mechanical condition, oil level for insulation and cooling, and ambient conditions if the hardware allows. It also normalizes local stress, calculates THI, evaluates local anomalies and risks, and sends alarms or triggers protection [3], [7], [13]. The main chip is an STM32F103, and it uses an ESP32/SX1278 LoRa transmitter for sending data.

B. Intelligent Gateway / Host Terminal

The host terminal, like a laptop or base computer, gets the LoRa data using an SX1278 receiver and a serial/JSON connection. It runs a Python app with a Streamlit dashboard showing live readings, historical trends, and predicting service dates based on wear. If the data shows a problem, the host creates a diagnostic question and sends it to a local large language model through Ollama [6]. It can also use a knowledge base of manufacturer and maintenance documents to help the model give a maintenance-focused explanation and a guide for technicians.

The work is divided between these two parts instead of being done all in one place for a few reasons. First, protection needs to work instantly, even if the communication or host software fails. This means the alarm and protection logic must run on the local microcontroller [5]. Second, tasks like visualization, trend analysis, and using AI are too much for the STM32, and are better suited for a more powerful host computer [3], [6]. Third, sending only a small, pre-processed JSON status instead of raw sensor data keeps the LoRa messages small, which is important because LoRa has a limited data rate [8].

V. SENSOR AND TELEMETRY ACQUISITION

The edge node gathers two types of measurements. One type looks at the transformer's own condition things like its electrical voltage, how much current it's drawing, its temperature, oil level, and vibrations [1], [3], [8]. The other type measures the environment around the transformer, such as the outside air temperature, humidity, and sometimes wind speed [12]. Table I shows all the sensors used in the test version.

Table I. Sensing Channels and Purpose

Parameter	Category	Sensing Element	Purpose
Voltage	Internal	Voltage-divider circuit	Over/under-voltage stress
Load current	Internal	Current transformer	Overload stress
Temperature	Internal	DS18B20 / LM35	Thermal stress, trip
Oil level	Internal	Float oil-level sensor	Insulation/cooling
Vibration	Internal	Piezoelectric sensor	Mechanical anomaly
Ambient temp.	External	Ambient temp. sensing	Environmental risk
Rel. humidity	External	Ambient humidity sensing	Environmental risk

Internal data helps calculate stress indicators and thermal equivalent loading, while external data informs the risk assessment [12], [13]. In the new system, the edge device collects both. It reads signals using its ADC and digital interfaces, standardizes each reading against reference values, and creates a single, compact JSON data package. This package includes both transformer health and environmental risk information, sent to the main system via LoRa [8]. The system uses only the sensors built into the prototype hardware or specified in the project design. If specific sensor models weren't provided, their function (like measuring temperature) is described instead of a model number.

VI. ELECTRICAL STRESS FINGERPRINTING AND TRANSFORMER HEALTH INDEX

Electrical Stress Fingerprinting (ESF) is key to the edge intelligence in both systems. It defines a transformer's normal operation using reference values for voltage, current, temperature, and vibration. When measured values differ from these references, a normalized stress indicator is calculated. Tracking these differences over time helps identify developing issues, rather than just reacting to immediate threshold breaches [3], [13]. Stress indicators are calculated as follows:

$$S_v = |V_{real} - V_n| / V_n, S_i = |I_{real} - I_n| / I_n \quad (1)$$

$$S_t = (T_{real} - T_n) / (T_{max} - T_n) \quad (2)$$

$$S_{vib} = (Vib_{real} - Vib_n) / (Vib_{max} - Vib_n) \quad (3)$$

The individual stress terms are aggregated into a single Transformer Health Index (THI) on a 0–100 scale:

$$THI = 100 - (W1S_v + W2S_i + W3S_t + W4S_{vib}) \times 100 \quad (4)$$

The weights (sum to 1) are example values, not optimized through experiments, and can be adjusted with more field data. THI offers a simple, single number for transformer condition, falling as stress increases [7], [13]. Analog measurements are converted to digital values using:

$$V(meas) = (ADC(value) / 4095) \times V(ref) \quad (5)$$

where ADC value is the digital reading and V ref is the reference voltage of the STM32 12-bit ADC. In the updated system, the ESF/THI calculation remains the same. Its output now feeds into the edge risk-fusion stage, along with the environmental state. This THI value is also sent over LoRa for remote visualization and AI analysis [8].

VII. REMAINING USEFUL LIFE AND DEGRADATION-BASED SERVICE PREDICTION

The simple, linear RUL estimate in ESF system is estimated based on the baseline with THI and the transformer's rated design life Lrated [13]:

$$RUL = L(rated) \times (THI / 100) \quad (6)$$

This is a simplified THI proportional relationship that is not based on a change in the rate of THI. This updated host software now also records the slope of the degradation process of THI over time. The daily degradation slope is where THI (initial) and THI (current) are recorded and t is the number of days elapsed between the two measurements.

$$m = (\text{THI}(\text{init}) - \text{THI}(\text{curr})) / \text{days} (7)$$

The host then utilises the observed slope to extrapolate ahead to when the observed THI will reach a set warning point (e.g. 70%) and generates a dynamically calculated Next Expected Service Date, not a calendar date. The distinction is highlighted explicitly in the paper: the proportional RUL value in (6) is retained as an estimate with low level of accuracy, which is a health-aware estimate, while the service-date-based extrapolation is treated as a health-insensitive operational scheduling aid, but whose accuracy is merely dependent on the length and quality of the historical THI record available at a given site, and which has not been validated against long-term failure data as reported in the provided project material [7], [9], [13] upon which it was built.

VIII. ENVIRONMENTAL-RISK FUSION

A transformer failure isn't always a problem confined to an individual item of equipment. If the ambient conditions at the transformer are sub-optimal (high ambient temperature, low relative humidity, dry vegetation and high wind), it is much more likely that an arcing failure, overheating or failure of the insulation to function, will lead to a vegetation fire or a larger fire in infrastructure than if the ambient conditions are cool, humid and quiet [12]. Current systems used to assess the health of the transformer equipment work on a per-equipment basis that do not consider this environmental context.

The updated architecture, then, incorporates an edge-level environmental-risk fusion stage for incorporating the transformer condition (calculated internally) and locally sensed environmental variables (Temperature, relative humidity and, where supported, wind-related variables) into a joint risk assessment [6], [10], [12]. This fusion isn't only used to make a statement that the transformer is in trouble, but to differentiate the chances of equipment failure from the severity of the resulting damage/disaster if that failure occurs with the prevailing local conditions.

The project material provided describes this component conceptually, in terms of an edge inference layer that is lightweight and couples equipment telemetry with fire-receptivity elements in question to yield a meteorological metric. This does not specify a trained model architecture, a labelled dataset or a measured classification accuracy. As with the non creation of un-backed technical detail, this paper hence presents environmental-risk fusion as a conceptual and architectural model realised as prototype logic on the edge controller, and not as an actual validated statistical or machine-learning model, with a measured accuracy value. Key future research in Section XVI mentions quantitative validation of output from risk-fusion with actual fire-weather and fault data.

IX. LORA POINT-TO-POINT COMMUNICATION

The transmitter and receiver are each an edge node and a host terminal that communicate via a point-to-point LoRa radio link with the help of an SX1278-type high gain transmitter. The edge node and host terminal use LoRa transceivers for communication over a point-to-point radio link. LoRa is a short term used for long range radio with low power, using a license free band, without requiring a cellular SIM, subscription to a carrier, or any

conventional network gateway making it an ideal choice for “rural” or “distributed” substations for various reasons, particularly for weak cell event coverage and low communication cost in the remote areas where substations are deployed.

The edge controller assembles a current-transformer state JSON telemetry (voltage, current, temperature, vibration, oil level, THI) payload and forward it to the host receiver directly and sends in the environmental-risk state. It's point-to-point, so there is no need for any intermediate infrastructure between the transformer site and the operator's location. On the host, the serial stream received is decotyped by the Python code controlling the Streamlit dashboard(s) described in Section XI. Focusing on fulfilling the command not to provide unverified figures, this paper does not present a specific communication range, packet-loss rate or latency, as these could not be measured in the presented project material; a quantification of how these metrics change affected by the field conditions is set as future work in [3], [4].

X. LOCAL EDGE-HOST AI DIAGNOSTIC ENGINE

A. Offline Operation

The first layer is a diagnostic layer that is installed and operates entirely on the host terminal using the Ollama local-inference runtime which enables running an open-source language model (e.g. a Phi-3-class or Llama-3-class model) without any call to the cloud or internet. This offline requirement also fits in perfectly with what we aim for with having a self-contained, rural-deployable system - without reliance on connectivity outside the local LoRa link [3].

B. Diagnostic Input

The AI layer will only be called upon if the host machine determines that a THI level has dropped below a specified warning value, indicating some kind of abnormal or degraded situation has occurred. The host then builds a question based on the relevant telemetry – THI and stress values, voltage, current, temperature, vibration, an oil level status, an environmental-risk state and a trend in these telemetry over the last, say, 10 minutes [3], [6].

C. Retrieval-Augmented Generation

Instead of directly using the general knowledge stored in the language model, the project matter consists of a description of the indexing of the relevant technical documents (manufacturer information, maintenance instructions, safety data, repair instructions) into a local library to retrieve. For diagnostic prompts and subsequent retrieval, the specific passages can be retrieved and fed into the model, which leads to a grounded explanation to the information that is relevant to the project instead of a nonconstrained information from language-model knowledge alone.

D. Diagnostic Output

The goal for the AI Layer is to provide a structured output including a likely cause explanation (which considers data quality and consistency), severity code, recommended inspection procedures and instrumentation and a technician-maintenance playbook [6]. This output is for the operator to use to speed up and standardise the operator's interpretation of an anomaly, and not to generate and override the protection commands.

This separation is intentional and made explicit: The AI-diagnostic layer is not for electrical safety reasons, but as a decision-support layer. The threshold to close the transformer relay, alarm to trip the switches and other safety-critical logic stay fully under control of the deterministic logic on the STM32 edge controller relying on sensors' logic described in Section VI; an AI response in absence, delayed or wrong: stays harmless. [5], [6].

XI. STREAMLIT OPERATOR DASHBOARD

The host-side operator interface is a Python/Streamlit dashboard that serves as the primary visualization layer of the updated system. Based on the supplied project material, the dashboard supports two acquisition modes a live mode that reads the STM32/LoRa serial stream, and an offline simulation mode used for demonstration and testing without live hardware. Supported display elements include a THI gauge, live voltage/current/temperature/vibration readouts, historical trend graphs, fault and anomaly alerts, and a dynamically projected next-expected-service date derived from the THI degradation slope described in Section VII [1], [3], [8], [13]. The dashboard also provides the operator-facing trigger for the AI diagnostic engine described in Section X, allowing the operator to request a detailed diagnostic explanation for a flagged anomaly [3], [6].

XII. MAINTENANCE DECISION-SUPPORT WORKFLOW

The end-to-end workflow linking sensing to maintenance action is summarized below:

- The edge node samples transformer and, where supported, environmental parameters [1], [4], [8], [12].
- The STM32 normalizes the measurements and computes the ESF stress indicators and THI (Section VI) [7], [9], [13].
- The edge risk-fusion stage evaluates whether local environmental conditions increase the consequence of any detected anomaly (Section VIII) [6], [10], [12].
- Critical electrical or thermal conditions trigger local alarms and, where thresholds are exceeded, deterministic relay-based load disconnection [5], [6].
- Compact JSON telemetry, including THI and the environmental-risk state, is transmitted to the host over the LoRa point-to-point link (Section IX) [3], [4].
- The Streamlit dashboard displays live values, trends, THI, environmental state and the projected next service date (Section XI) [1], [3], [8], [13].
- When an anomaly requires further explanation, the host constructs a structured prompt for the offline AI diagnostic engine (Section X) [3], [6].
- The AI engine retrieves relevant technical guidance and returns a maintenance-oriented diagnostic explanation and technician playbook [6].
- The operator combines the generated explanation with standard engineering judgement and procedure to plan inspection and dispatch.

As emphasized throughout Sections VI and X, the AI-assisted steps in this workflow operate strictly as decision support; the protective actions earlier in the workflow are deterministic and do not depend on the availability of the host, the LoRa link or the AI service [5], [6].

XIII. PROTOTYPE IMPLEMENTATION AND RESULTS

The hardware validation reported in this section is carried over from the baseline ESF prototype and provides the physical foundation on which the updated LoRa/environmental/AI architecture is built [13]. Fig. 2 shows the block-level design of the sensing, processing and alert/protection chain, and Fig. 3 shows the underlying circuit implementation around the STM32F103C8T6 microcontroller. The system is powered from a 230 V AC supply, stepped down and rectified to regulated 12 V and 5 V rails; voltage is sensed through a divider/rectifier network, load current through a current transformer, temperature through a DS18B20/LM35 sensor, and vibration through a piezoelectric sensor, with all channels digitized by the STM32 internal ADC and displayed on a 16×2 LCD [1], [4], [8], [13].

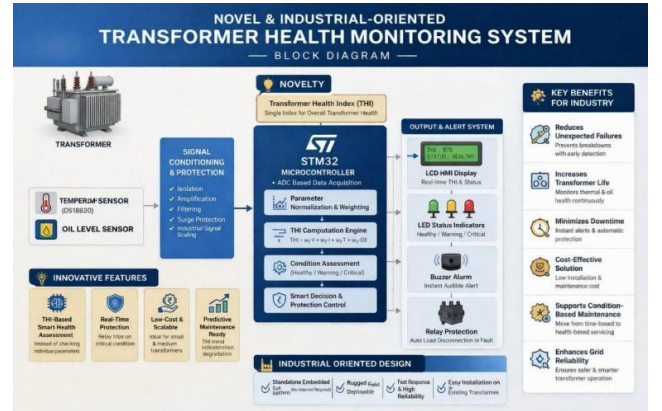


Fig. 2. Block-level design of the STM32-based transformer health-monitoring and protection chain (baseline hardware foundation).

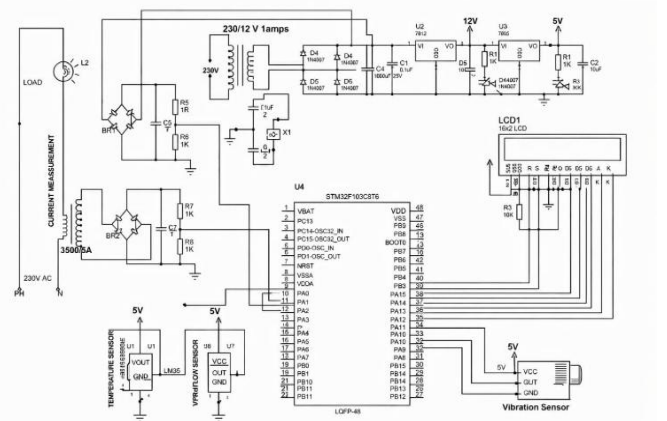


Fig. 3. Circuit implementation of the STM32F103C8T6-based sensing, power-supply and protection stage.

Fig. 4 shows the assembled prototype test setup, including the step-down transformer, current transformer, relay module, signal-conditioning stage, STM32 module, LCD display, buzzer, LED indicators, vibration sensor, oil-level sensor and temperature sensor mounted on a test board with resistive load (bulbs) and a manually operated fault-creation unit used to induce overload and other test conditions [1], [4], [8], [13].

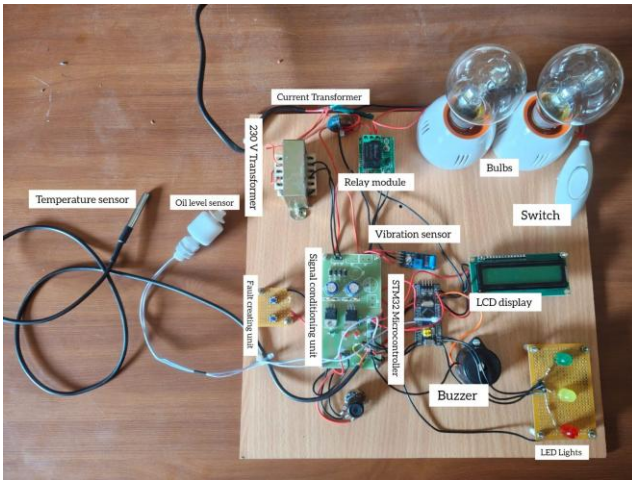


Fig. 4. Assembled prototype hardware test setup.

Under a representative stressed test condition, the LCD reported approximately 336 V, 79 A load current, 35 °C and a computed THI of 59, reflecting the reduced health index expected under an overload scenario (Fig. 5) [13]. Corresponding LED and buzzer alerts were activated. In separate test cases, the prototype also produced a vibration alert when mechanical disturbance exceeded the configured threshold, a high-temperature trip when winding temperature exceeded the safe operating limit, a low-oil-level indication when the sensed oil level fell below the configured limit, and an overload trip with automatic relay disconnection when load current exceeded the rated limit [1], [4], [8], [13]. These results validate the sensing, THI computation, alerting and protection functions on which the updated LoRa/environmental/AI architecture is built, and should be read as prototype-scale observations rather than a statistically characterized field-performance evaluation [13].

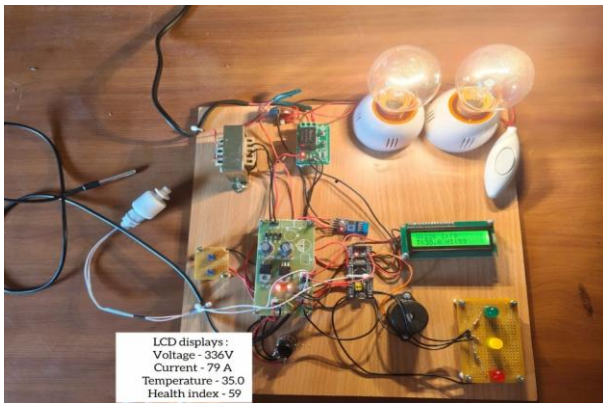


Fig. 5. Representative prototype display showing voltage, current, temperature and computed THI under a stressed load condition.

The LoRa telemetry link, Streamlit dashboard and offline AI diagnostic layer described in Sections IX–XI extend this validated hardware foundation architecturally [3], [6]. Consistent with the requirement to avoid unverified claims, this paper does not report a measured LoRa communication range, AI diagnostic accuracy, response latency or fire-risk classification accuracy, since these quantities were not experimentally measured in the supplied project material. Where the updated project material references such figures only as literature-reported values, they are treated in this

paper as background context rather than as results of the present system.

XV. DISCUSSION

The integrated design addresses the three gaps identified in Section II. First, small and remote distribution transformers often cannot justify the cost and connectivity requirements of SCADA and continuous DGA deployments; the proposed edge node uses commodity microcontroller and sensor hardware and communicates over a license-free, gateway-free LoRa link, which is intended to substantially reduce both capital and recurring communication cost relative to cellular- or SCADA-based alternatives [2], [3], [4], [8]. Second, transformer condition and environmental fire risk are commonly assessed by separate systems; the edge risk-fusion stage in Section VIII couples them at the point of measurement, allowing maintenance priority to reflect not only equipment health but also the local consequence of a potential fault [6], [10], [12]. Third, raw telemetry is often difficult for a field operator to convert into a concrete maintenance action; the offline, retrieval-grounded AI diagnostic layer in Section X is intended to translate anomalous telemetry into a structured, technician-usable explanation without requiring cloud connectivity [3], [6].

The architecture also supports incremental deployment. Because the edge controller retains full sensing, THI computation and protection functionality independent of the host, a site can be commissioned first as a standalone health monitor using the validated baseline hardware, then extended with the LoRa transmitter and environmental sensing, and finally connected to the Streamlit host and offline AI layer as those components are brought online. This staged approach limits integration risk, since the safety-critical monitoring and protection functions remain operational throughout the rollout of the higher-level software components [1], [3], [4], [13].

XVI. LIMITATIONS

- The reported hardware results are prototype-scale observations on a bench test rig with a resistive load and manually induced faults, rather than field measurements on an in service distribution transformer [13].
- The THI weighting coefficients (W1–W4) are illustrative design values rather than coefficients derived from a statistically fitted or experimentally optimized model, and require calibration against a larger transformer dataset [7], [9], [13].
- The environmental-risk-fusion framework is described conceptually in this paper; it has not yet been validated against a labelled dataset of transformer faults and corresponding weather/fire outcomes [6], [12].
- The RUL and degradation-based service-date estimates are derived from THI trends rather than from long-term historical failure or degradation records, and their predictive accuracy has not been independently validated [7], [9], [13].

LoRa communication range, packet loss and latency were not experimentally characterized in the supplied project material and are expected to depend strongly on the deployment terrain and antenna configuration.

- AI-generated diagnostic output requires verification by a qualified technician or engineer before acting on it; it is a decision-support aid and not an authoritative fault determination [6].
- The system has not undergone long-term field testing under real seasonal, loading and weather variation [13].

XVII. FUTURE WORK

- Collect a larger multi-transformer dataset spanning varied loading, ambient and seasonal conditions to calibrate THI weights and validate RUL/service-date estimates [7], [9], [13].
- Quantitatively evaluate LoRa range, packet-loss rate, latency and energy consumption under representative rural-substation deployment conditions.
- Validate the environmental-risk-fusion framework against recorded fault and weather/fire outcome data, potentially using a lightweight (TinyML-class) on-device model [6], [12].
- Add solar-powered supply, battery backup and low-power sleep modes to support autonomous, self-sustaining edge-node operation.
- Benchmark candidate offline language models for diagnostic consistency, response latency and hallucination resistance [6].
- Capture technician inspection outcomes as feedback to continuously improve the anomaly and risk-fusion models.
- Develop secure interfaces for optional integration with utility SCADA or control-room systems [1], [3], [4].

XVIII. CONCLUSION

This paper has presented the evolution of a low-cost STM32-based Electrical Stress Fingerprinting transformer health and life-prediction system into a decentralized Intelligent Grid Network [13]. The updated architecture retains the validated multi-sensor acquisition, THI computation, life estimation, alarm and relay-protection functions of the original system, and extends them with edge-level environmental-risk fusion, point-to-point LoRa telemetry that removes dependence on cellular infrastructure, a Streamlit-based operator dashboard with degradation-based service-date projection, and an offline, retrieval-grounded AI diagnostic layer for maintenance decision support [1], [3], [6], [8], [10], [12]. The guiding design principle throughout is split intelligence: safety-critical sensing and protection remain deterministic and local to the transformer, while the host terminal performs visualization, trend analysis and generative diagnostic reasoning [3], [5], [6]. The reported prototype hardware results validate the underlying sensing, health-index and protection platform, while the environmental-risk fusion, LoRa telemetry, dashboard and AI diagnostic layer are presented as an integrated architecture and prototype-level capability consistent with the supplied project material, rather than as field-proven or production-ready subsystems [13]. Future work should focus on quantitative validation of the environmental-risk model, characterization of LoRa performance in the field, benchmarking of the offline AI diagnostic layer, and long-term, multi-transformer field deployment to establish the accuracy of life and service-date prediction [6], [7], [9], [12], [13].

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